Day in the life of

Exercise
- Context
- The family
- The task
- The API
- Requirements
- Ground rules

Exercise

Time	90 minutes
Format	In-office, your own machine, any tools you normally use
Stack	Your choice; whatever makes you go fast
Timeline	• Exercise is shared and you can read it (10min) • Pair Q&A/planning session w/ Nick (10min) • Execute (70min)

Context

Gas prices are back in the news. With tensions rising in the Middle East, Belgian households are wondering what they actually paid for gas over the past months, and whether prices moved with the news cycle.

We'd like to make it easy to answer this question for the "*De Smedt family*". Today, you'll build a small prototype that does exactly this for the family.

The family

The De Smedt family lives in Ghent (9000). They have a residential gas connection. Their monthly gas consumption over the past 7 months was:

Month	Consumption
October 2025	1,000 kWh

Month	Consumption
November 2025	1,200 kWh
December 2025	2,500 kWh
January 2026	2,800 kWh
February 2026	2,500 kWh
March 2026	1,500 kWh
April 2026*	1,000 kWh

** for the purpose of this exercise you can assume April 2026 has passed and total consumption for that month was 1,000 kWh.*

During this period, the family had the “Eneco Aardgas Flex” tariff (variabel).

The task

Build a small app that shows the De Smedt family **what their gas cost them each month**, and how that evolved over the 7-month period.

The output is a visualization (chart, table, something else...) with at least three things on it:

- Their **monthly consumption** (kWh)
- Their **monthly cost** (€ incl. VAT)
- Cost-per-kWh ratio (€c/kWh)

The goal is to make the relationship between the three visible — and specifically to surface whether the cost-per-kWh moved over time, not just the total bill.

That's the core.

As a potential next step, if time allows it, add a dropdown that lets the De Smedt family compare their Eneco contract against any other **variable gas** contract in the market. Re-run the simulations for all months for the selected product and make it easy to see whether they're (and have been) on a good deal.

The API

You have access to **Vwala’s Open Energie API**.

API Documentation:
https://open-energie.docs.vwala.be/

You can generate an API key by following the four steps outlined in the auth guide.

Requirements

The only hard requirement is that your setup is a full-stack app with a back-end and front-end. You can choose the stack but one example framework is NextJS which allows building a front- and back-end in the same app (another example: FastAPI as back-end with Jupyter notebook as front-end; you get the point...).

The backend:
Exposes an endpoint that can run the simulations and return the results in a shape your frontend can display. Your back-end is thus responsible for calling the Open Energie API.

A frontend (or at least something that allows us to display the results)

- A chart or other visualization that plots the monthly consumption vs. monthly cost over the given time period
- The De Smedt family's data pre-loaded — no need for a user input form

Ground rules

- Use any tools you normally use: Cursor, Claude, Copilot, whatever.
- It's fine to cut scope. A focused, working thing is better than an ambitious broken one.
- If something is ambiguous, make a decision, leave a comment explaining why, and move on.